

x62-data-challenge-student-pathways

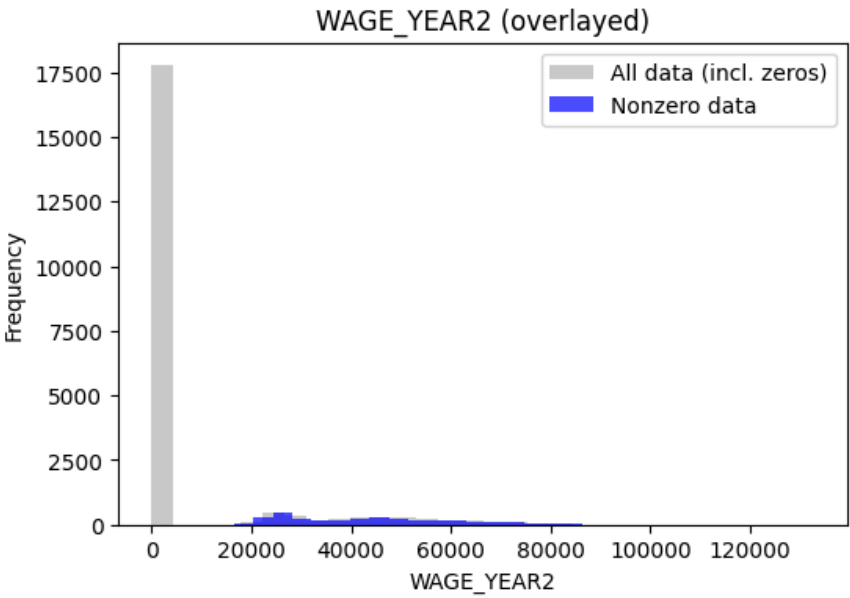
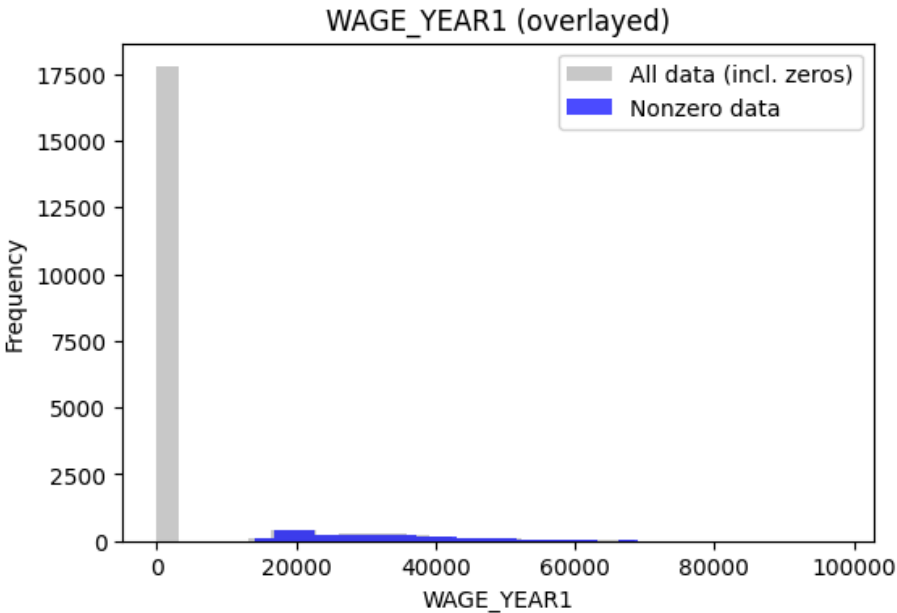
1) Data Quality:-----

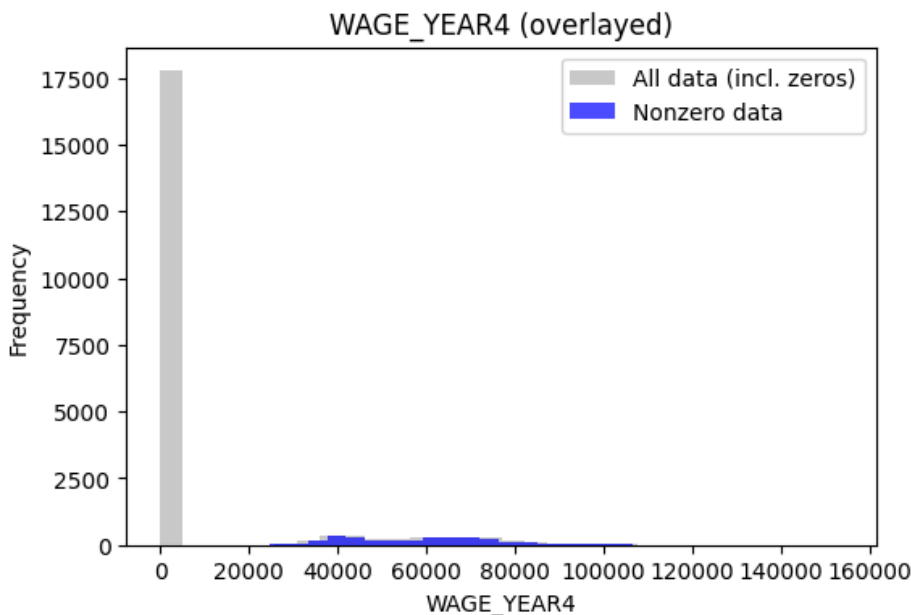
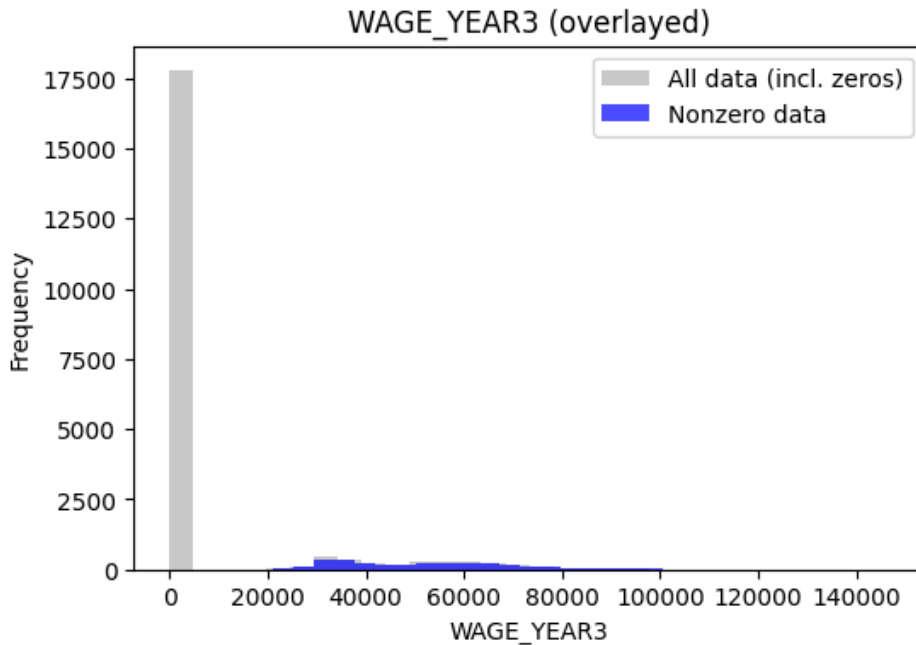
Initially, my data quality check suggested that only DISTRICT_CODE had missing values as seen in the print outs below. I used histograms and summary statistics (as seen below) to visualize the distribution of wage values and quickly noticed large spikes at zero, confirming this issue. After accounting for zeros as missing, I identified 2,745 missing DISTRICT_CODE entries and about 17,770 missing wage values in the training set, along with 907 missing DISTRICT_CODE and 5,917 missing wage entries in the test set. This deeper analysis revealed that while most categorical features are complete and reliable, the wage data contains significant gaps that must be cleaned and handled properly to ensure accurate and unbiased modeling results.

Data Types (df): DISTRICT_TYPE object DISTRICT_NAME object DISTRICT_CODE float64 ACADEMIC_YEAR object DEMO_CATEGORY object STUDENT_POPULATI ON object AWARD_CATEGORY object WAGE_YEAR1 float64 WAGE_YEAR2 float64 WAGE_YEAR3 float64 WAGE_YEAR4 float64 dtype: object	Missing Data (including 0s) in df: DISTRICT_TYPE 0 DISTRICT_NAME 0 DISTRICT_CODE 2745 ACADEMIC_YEAR 0 DEMO_CATEGORY 0 STUDENT_POPULATI ON 0 AWARD_CATEGORY 0 WAGE_YEAR1 17770 WAGE_YEAR2 17770 WAGE_YEAR3 17770 WAGE_YEAR4 17770 dtype: int64	Data Types (df_predict): DISTRICT_TYPE object DISTRICT_NAME object DISTRICT_CODE float64 ACADEMIC_YEAR object DEMO_CATEGORY object STUDENT_POPULATI ON object AWARD_CATEGORY object WAGE_YEAR1 float64 WAGE_YEAR2 float64 WAGE_YEAR3 float64 dtype: object	Missing Data (including 0s) in df_predict: DISTRICT_TYPE 0 DISTRICT_NAME 0 DISTRICT_CODE 907 ACADEMIC_YEAR 0 DEMO_CATEGORY 0 STUDENT_POPULATI ON 0 AWARD_CATEGORY 0 WAGE_YEAR1 5917 WAGE_YEAR2 5917 WAGE_YEAR3 5917 dtype: int64
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	DISTRICT_CODE	WAGE_YEAR1	WAGE_YEAR2	WAGE_YEAR3	WAGE_YEAR4
count	1.796000e+04	20705.000000	20705.000000	20705.000000	20705.000000
mean	3.041331e+06	4476.106834	6075.533253	7310.831635	8530.890413
std	1.583286e+06	11944.502346	16140.916903	19158.203471	22106.663179
min	1.100170e+05	0.000000	0.000000	0.000000	0.000000
25%	1.864089e+06	0.000000	0.000000	0.000000	0.000000

50%	3.166852e+06	0.000000	0.000000	0.000000	0.000000
75%	4.277214e+06	0.000000	0.000000	0.000000	0.000000
max	5.872769e+06	97993.000000	132847.000000	146728.000000	153910.000000





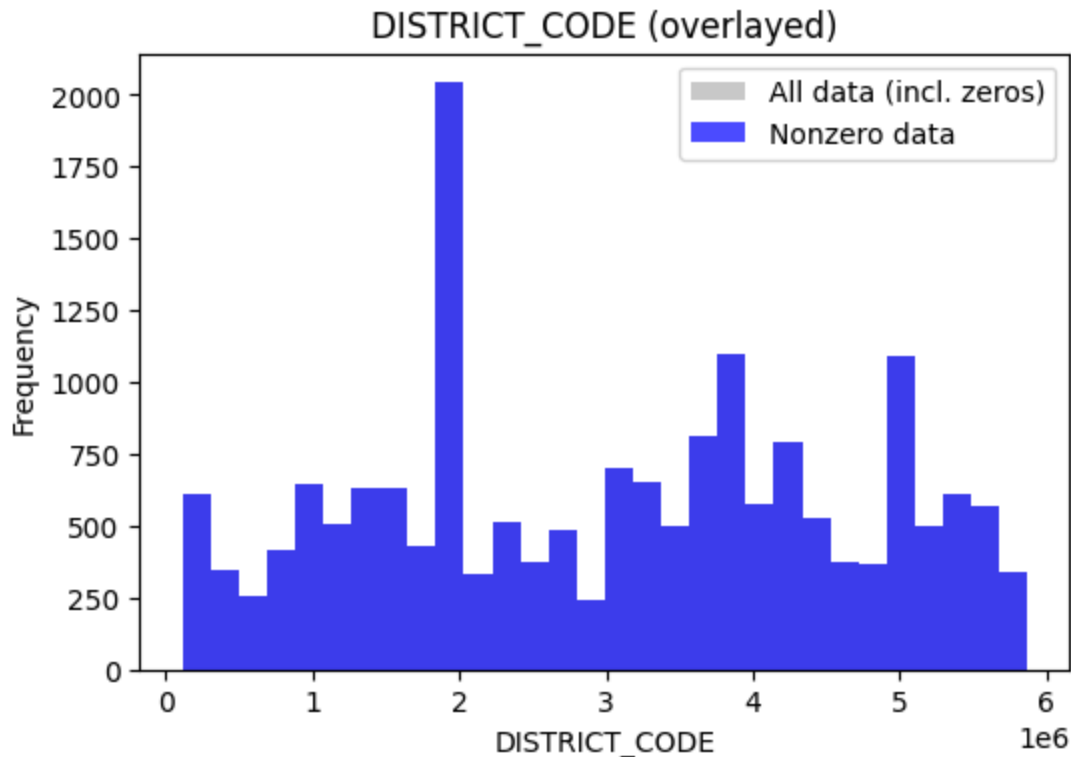
2) Range: -----

The range analysis as seen below showed that categorical columns had limited unique values. DISTRICT_TYPE included three groups (School District, Legislative District, All), DEMO_CATEGORY had five (such as Race and Gender), and STUDENT_POPULATION contained multiple subgroups like Hispanic or Latino and Foster Youth. ACADEMIC_YEAR had only 2018–2019, and AWARD_CATEGORY included four degree types. For numeric columns, DISTRICT_CODE ranged from about 110,000 to 5.8 million, showing slight spikes around 2, 4, and 5 in the graph below. From the Graphs (pictured above) we see that the wage columns (WAGE_YEAR1–4) ranged from 0 to roughly 150,000 and were all right-skewed. The vast majority were 0 but most paid

salaries were between 20,000 and 40,000, but as the years go by a second spike starts to grow around the 60,000 range.

feature: DISTRICT_TYPE ['School District' 'Legislative District' 'All']	feature: ACADEMIC_YEAR ['2018-2019']	feature: DEMO_CATEGORY ['Race' 'Homeless Status' 'All' 'Foster Status' 'Gender']
feature: DISTRICT_NAME ['Duarte Unified' 'Coronado Unified' 'Gilroy Unified' 'Pleasant Valley' 'Senate District 15' 'Adelanto Elementary' 'Assembly District 56' ..(and more)... 'North Monterey County Unified' 'Fontana Unified' 'Assembly District 5' 'Redlands Unified']	feature: STUDENT_POPULATION ['None Reported' 'Black or African American' 'Did Not Experience Homelessness in K-12' 'American Indian or Alaska Native' 'Native Hawaiian or Other Pacific Islander' 'All' 'Two or More Races' 'Foster Youth' 'Female' 'White' 'Experienced Homelessness in K-12' 'Not Foster Youth' 'Male' 'Asian' 'Hispanic or Latino']	feature: AWARD_CATEGORY ['Bachelor's Degree - Did Not Transfer' 'Associate Degree' 'Community College Certificate' 'Bachelor's Degree - Transferred']

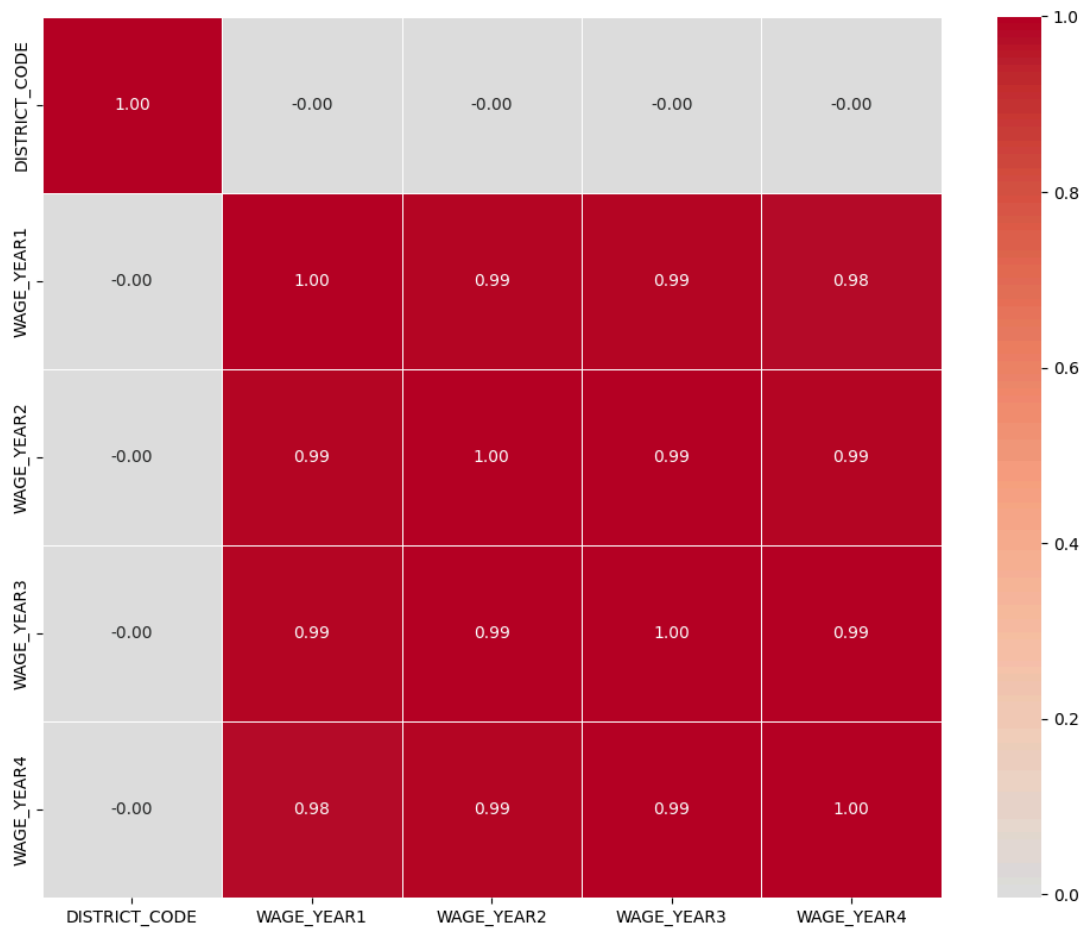
	DISTRICT_CODE	WAGE_YEAR1	WAGE_YEAR2	WAGE_YEAR3	WAGE_YEAR4
count	1.796000e+04	20705.000000	20705.000000	20705.000000	20705.000000
mean	3.041331e+06	4476.106834	6075.533253	7310.831635	8530.890413
std	1.583286e+06	11944.502346	16140.916903	19158.203471	22106.663179
min	1.100170e+05	0.000000	0.000000	0.000000	0.000000
25%	1.864089e+06	0.000000	0.000000	0.000000	0.000000
50%	3.166852e+06	0.000000	0.000000	0.000000	0.000000
75%	4.277214e+06	0.000000	0.000000	0.000000	0.000000
max	5.872769e+06	97993.000000	132847.000000	146728.000000	153910.000000

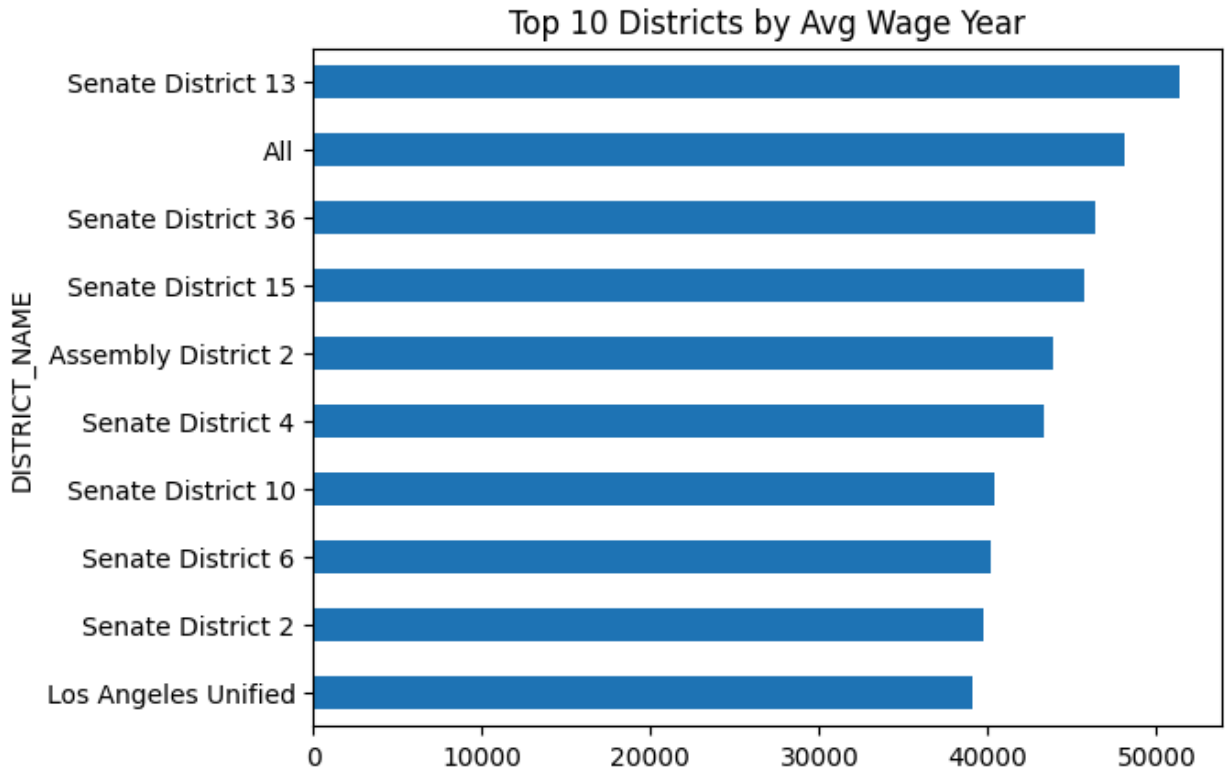


3) Semantics: -----

Each column provides specific contextual information about district characteristics, demographics, and wage outcomes. DISTRICT_TYPE and DISTRICT_NAME identify the organization and location, while DISTRICT_CODE serves as a numeric identifier. ACADEMIC_YEAR shows the time frame of the data (2018–2019), and DEMO_CATEGORY along with STUDENT_POPULATION describe demographic breakdowns like race, gender, and foster or homelessness status. AWARD_CATEGORY represents educational outcomes, such as degrees or certificates earned.

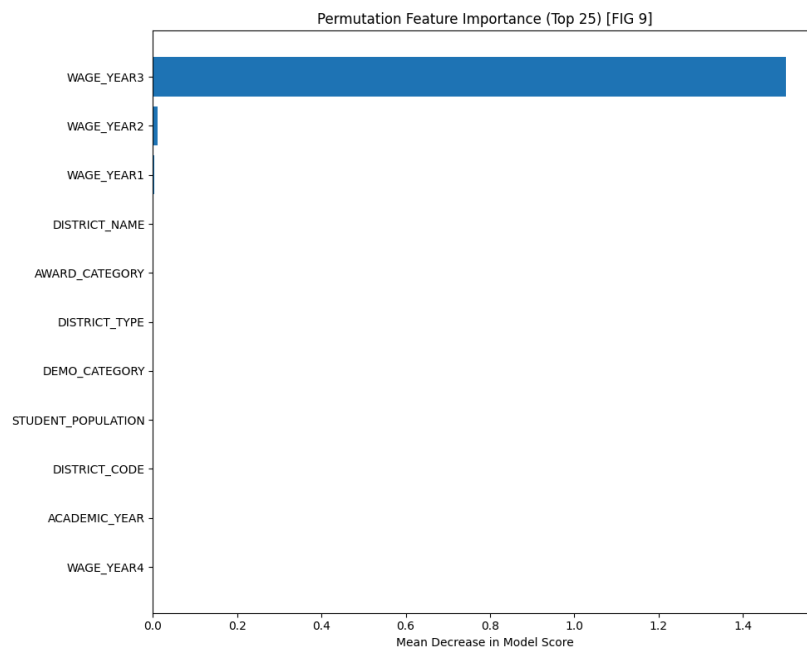
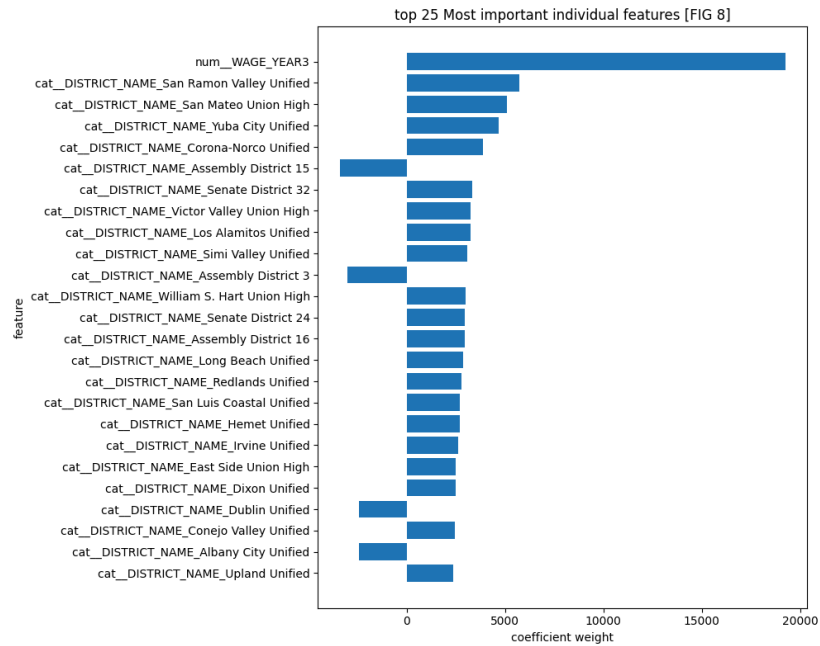
From the heatmap (shown below), I found that DISTRICT_CODE has no correlation with any wage columns, confirming it functions as an identifier rather than a predictive factor. However, Wage_Year1 through Wage_Year4 are almost perfectly correlated (around 0.99), which I visualized using the correlation heatmap. This means wage outcomes across consecutive years are highly related, suggesting strong year-over-year consistency in earnings data and potential redundancy among those features. However, there does seem to be a strong correlation between Wage_Year and the districts as can be seen in the histogram below showing top districts with highest pay.

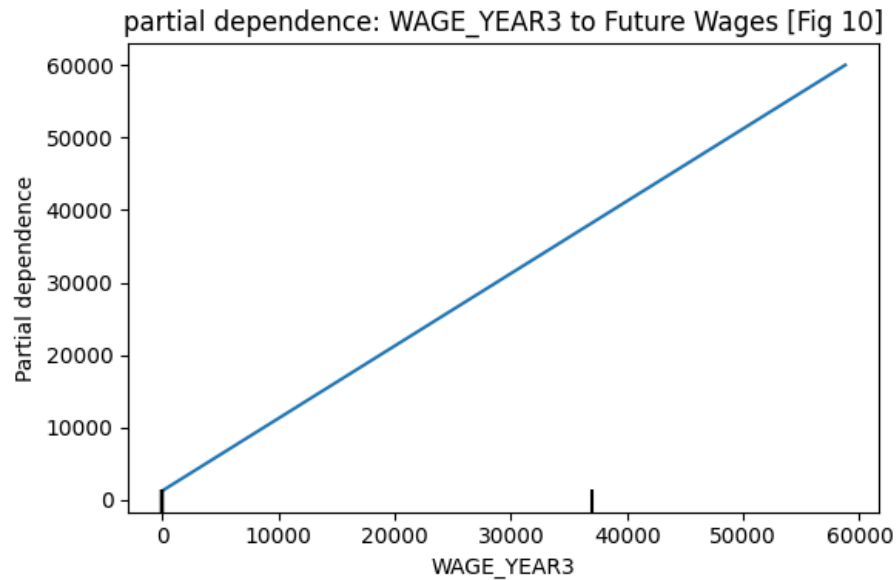




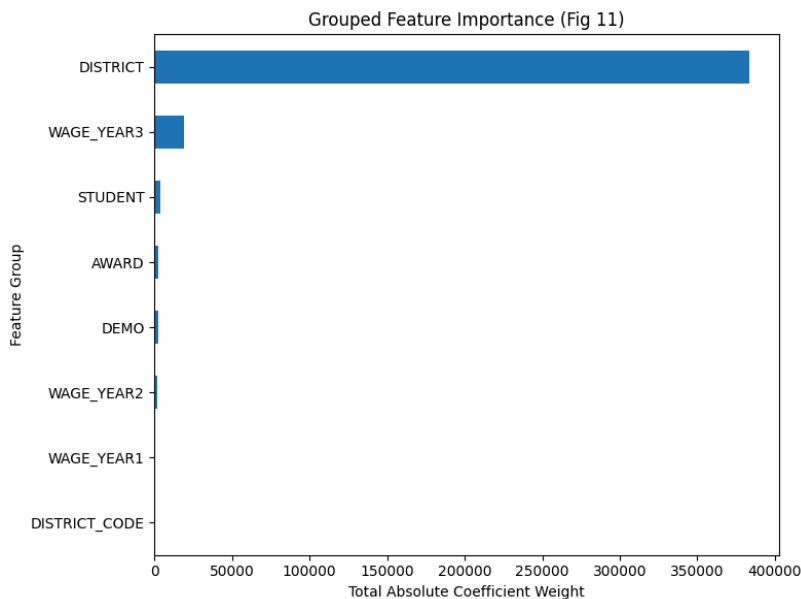
Reflection: -----

1) These results show that Wage_Year3 is the strongest predictor of future earnings, which makes sense since past wages usually reflect future income. District-level variables also matter, meaning where someone works can affect their salary. To prove this, I created two visualizations: the Top-25 Coefficient Plot (Fig. 8), which highlights the biggest model weights, and the Permutation Feature Importance (Fig. 9), which measures how much performance drops when each feature is shuffled. Both clearly show that Wage_Year3 has the greatest impact. I also show the partial dependence between Wage_Year3 to future wages as seen in (Fig. 10) also below.





2) When we look beyond Wage_Year3, the district someone works in clearly has a big influence on their future wages. This makes sense because different districts have different funding levels, cost of living, and salary structures. In the earlier plots that I made, individual districts didn't seem as important because each one was a one hot encoded district column spread out across hundreds of tiny features. But when we group all district dummy variables together, you can see just how impactful the district really is. The grouped feature importance plot (Fig. 11) below shows that district-level effects are one of the strongest predictors overall.



3) Some features that might shed more light would be things like the district/school yearly budget. Maybe things like the student tax bracket might also tell us how much tax money is funding these schools affecting wages. job field, or where they live might also help. Stuff like that could help the model understand why some people earn more than others instead of just guessing from the limited data we have.